THE ESSENTIAL REVIEW

Grade 10 General

مراجعة مركّزة من الكتاب

Exponents. Polynomials. Quadratics.

25 learning outcomes

101 textbook questions

60 minute online mock

Name / الاسم _____ Class / الصف _____

A short route to confident practice

Read the rule → solve without looking at the key → check your steps → repeat the topic online.

اقرأ القاعدة، حلّ بنفسك، ثم راجع خطواتك. لكل مخرج 4 أسئلة مختارة، ولخواص ضرب القوى 5 أسئلة لتغطية الأفكار المتعددة.

Your topic map

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Selected from the uploaded Reveal textbook, printed pp. 569–738, using the supplied 2026–2027 Grade 10 General coverage. Numbers and diagrams come from the book; MCQ choices and worked solutions are added. This is a study selection, not an official exam or a prediction. The official coverage allows a scientific calculator and specifies 150 minutes / 100 marks. The website mock is a separate 60-minute practice format.

الأسئلة من الكتاب فقط، مع توثيق الصفحة والتمرين. لا يمكن ضمان ورود أسئلة بعينها في الاختبار النهائي.

01 / CHOOSE & CHECK

Q1 — Q2

Q1 FOIL binomial products

ضرب ثنائيي الحد

Remember Multiply all four pairs, then combine the two middle terms.

1.1 MCQ

Find the product.

$$(c+3)(3c-5)$$

Reveal · p. 649 · 1 · MCQ adaptation

A $3c^2+3c-15$ B $3c^2+4c-14$

C $3c^2+4c-15$ D $3c^2+5c-15$

1.2 MCQ

Find the product.

$$(g+10)(2g-5)$$

Reveal · p. 649 · 2 · MCQ adaptation

A $2g^2+15g-49$ B $2g^2+15g-50$

C $2g^2+16g-50$ D $2g^2+14g-50$

1.3 MCQ

Find the product.

$$(5a+3)(6a+5)$$

Reveal · p. 649 · 3 · MCQ adaptation

A $30a^2+43a+15$ B $30a^2+44a+15$

C $30a^2+42a+15$ D $30a^2+43a+16$

1.4 MCQ

Find the product.

$$(4x+1)(6x+3)$$

Reveal · p. 649 · 4 · MCQ adaptation

A $24x^2+19x+3$ B $24x^2+17x+3$

C $24x^2+18x+4$ D $24x^2+18x+3$

Q2 Squares of sums and differences

مربع مجموع وفرق

Remember $(a \pm b)^2 = a^2 \pm 2ab + b^2$. Do not omit the middle term.

2.1 MCQ

Find the product.

$$(a+10)(a+10)$$

Reveal · p. 657 · 1 · MCQ adaptation

A $a^2+20a+101$ B $a^2+20a+100$

C $a^2+21a+100$ D $a^2+19a+100$

2.2 MCQ

Find the product.

$$(b-6)(b-6)$$

Reveal · p. 657 · 2 · MCQ adaptation

A $b^2-12b+36$ B $b^2-11b+36$

C $b^2-13b+36$ D $b^2-12b+37$

2.3 MCQ

Find the product.

$$(n+7)^2$$

Reveal · p. 657 · 3 · MCQ adaptation

A $n^2+15n+49$ B $n^2+13n+49$

C $n^2+14n+50$ D $n^2+14n+49$

2.4 MCQ

Find the product.

$$(x+6)^2$$

Reveal · p. 657 · 4 · MCQ adaptation

A $x^2+11x+36$ B $x^2+12x+37$

C $x^2+12x+36$ D $x^2+13x+36$

01 / CHOOSE & CHECK

Q3 — Q4

Q3 Sum and difference products

حاصل ضرب مجموع وفرق

Remember (a+b)(a−b)=a²−b²; the middle terms cancel.

3.1 MCQ

Find the product.

$$(u+3)(u-3)$$

Reveal · p. 658 · 15 · MCQ adaptation

A u^2-9
B u^2+9
C u^2+3u-9
D $9-u^2$

3.2 MCQ

Find the product.

$$(b+7)(b-7)$$

Reveal · p. 658 · 16 · MCQ adaptation

A b^2+49
B $b^2+7b-49$
C $49-b^2$
D b^2-49

3.3 MCQ

Find the product.

$$(2+x)(2-x)$$

Reveal · p. 658 · 17 · MCQ adaptation

A $-x^2+2x+4$
B x^2-4
C $4-x^2$
D x^2+4

3.4 MCQ

Find the product.

$$(4+x)(4-x)$$

Reveal · p. 658 · 18 · MCQ adaptation

A x^2-16
B $16-x^2$
C x^2+16
D $-x^2+4x+16$

Q4 Factor using distribution

التحليل بخاصية التوزيع

Remember Take the greatest common factor from every term; check by multiplication.

4.1 MCQ

Factor completely over the integers.

$$16t-40y$$

Reveal · p. 665 · 1 · MCQ adaptation

A $\frac{(-1)8(-2t+5y)}{2}$
B $2\left(-8\left(-2t+5y\right)\right)$
C $8\left(-2t+5y\right)$
D $-8\left(-2t+5y\right)$

4.2 MCQ

Factor completely over the integers.

$$30v+50x$$

Reveal · p. 665 · 2 · MCQ adaptation

A $2\cdot 10\left(3v+5x\right)$
B $-10\left(3v+5x\right)$
C $10\left(3v+5x\right)$
D $\frac{10(3v+5x)}{2}$

4.3 MCQ

Factor completely over the integers.

$$2k^2+4k$$

Reveal · p. 665 · 3 · MCQ adaptation

A $-2k\left(k+2\right)$
B $2k\left(k+2\right)$
C $\frac{2k(k+2)}{2}$
D $2\cdot 2k\left(k+2\right)$

4.4 MCQ

Factor completely over the integers.

$$5z^2+10z$$

Reveal · p. 665 · 4 · MCQ adaptation

A $5z\left(z+2\right)$
B $\frac{5z(z+2)}{2}$
C $2\cdot 5z\left(z+2\right)$
D $-5z\left(z+2\right)$

01 / CHOOSE & CHECK

Q5 — Q6

Q5 Analyse monic trinomial factors

تحليل ثلاثيات الحدود ومعاملاتها

Remember For (x+r)(x+s), r+s=b and rs=c. Enumerate integer factor pairs.

5.2 MCQ

Jamaall: (x+2)(x−8). Charles: (x−2)(x+8). Who correctly factored x²+6x−16?

Reveal · p. 674 · 74 · MCQ adaptation

A Neither student
B Charles only
C Jamaall only
D Both students

5.3 MCQ

Find all values of k for which the polynomial factors using integers.

$$x^2 + kx - 19$$

Reveal · p. 674 · 75 · MCQ adaptation

A $k \in \{-18, 18\}$
B $k \in \{-20, 20\}$
C $k \in \{-19, 19\}$
D $k = 18$

5.5 MCQ

Find all values of k for which the polynomial factors using integers.

$$x^2 - 8x + k, \quad k > 0$$

Reveal · p. 674 · 77 · MCQ adaptation

A $k \in \{8, 16\}$
B $k \in \{1, 2, 3, 4\}$
C $k \in \{7, 12, 15, 16\}$
D $k \in \{7, 12, 15\}$

5.6 MCQ

Find all values of k for which the polynomial factors using integers.

$$x^2 - 5x + k, \quad k > 0$$

Reveal · p. 674 · 78 · MCQ adaptation

A $k = 6$
B $k \in \{4, 6\}$
C $k \in \{4, 5, 6\}$
D $k \in \{1, 5\}$

Q6 Factor non-monic trinomials

تحليل ثلاثيات الحدود غير الأحادية

Remember Find two numbers with product ac and sum b; split the middle term and group.

6.1 MCQ

Factor completely over the integers.

$$5x^2 + 34x + 24$$

Reveal · p. 672 · 21 · MCQ adaptation

A $2\,(x + 6)\,(5x + 4)$
B $(x + 6)\,(5x + 4)$
C $(x + 4)\,(5x + 6)$
D $-\,(x + 6)\,(5x + 4)$

6.2 MCQ

Factor completely over the integers.

$$2x^2 + 19x + 24$$

Reveal · p. 672 · 22 · MCQ adaptation

A $(x + 8)\,(2x + 3)$
B $(x + 3)\,(2x + 8)$
C $-\,(x + 8)\,(2x + 3)$
D $2\,(x + 8)\,(2x + 3)$

6.3 MCQ

Factor completely over the integers.

$$4x^2 + 22x + 10$$

Reveal · p. 672 · 23 · MCQ adaptation

A $(2x + 2)\,(2x + 5)$
B $-2\,(x + 5)\,(2x + 1)$
C $4\,(x + 5)\,(2x + 1)$
D $2\,(x + 5)\,(2x + 1)$

6.4 MCQ

Factor completely over the integers.

$$4x^2 + 38x + 70$$

Reveal · p. 672 · 24 · MCQ adaptation

A $-2\,(x + 7)\,(2x + 5)$
B $4\,(x + 7)\,(2x + 5)$
C $2\,(x + 7)\,(2x + 5)$
D $(2x + 7)\,(2x + 10)$

01 / CHOOSE & CHECK

Q7

Q7 Difference of squares

تحليل الفرق بين مربعين

Remember $A^2-B^2=(A-B)(A+B)$. Factor repeatedly when possible.

7.1 MCQ

Factor completely over the integers.

$$q^2-121$$

Reveal · p. 679 · 1 · MCQ adaptation

A $(q-11)(q+11)$

B $2(q-11)(q+11)$

C $\frac{(q-11)(q+11)}{2}$

D $-(q-11)(q+11)$

7.2 MCQ

Factor completely over the integers.

$$-k^4+r^4$$

Reveal · p. 679 · 2 · MCQ adaptation

A $2(-k+r)(k+r)\left(k^2+r^2\right)$

B $\frac{(-k+r)(k+r)\left(k^2+r^2\right)}{2}$

C $-(-k+r)(k+r)\left(k^2+r^2\right)$

D $(-k+r)(k+r)\left(k^2+r^2\right)$

7.3 MCQ

Factor completely over the integers.

$$w^4-625$$

Reveal · p. 679 · 3 · MCQ adaptation

A $\frac{(w-5)(w+5)\left(w^2+25\right)}{2}$

B $-(w-5)(w+5)\left(w^2+25\right)$

C $(w-5)(w+5)\left(w^2+25\right)$

D $2(w-5)(w+5)\left(w^2+25\right)$

7.4 MCQ

Factor completely over the integers.

$$r^2-9t^2$$

Reveal · p. 679 · 4 · MCQ adaptation

A $-(r-3t)(r+3t)$

B $(r-3t)(r+3t)$

C $2(r-3t)(r+3t)$

D $\frac{(r-3t)(r+3t)}{2}$

01 / CHOOSE & CHECK

Q8

Q8

Divide monomials

قسمة وحيدات الحد

Remember Subtract exponents when dividing; multiply every exponent when raising a quotient to a power.

8.2 MCQ

Simplify. Assume no denominator equals zero.

$$\left(\frac{p^2t^7}{10}\right)^3$$

Reveal · p. 583 · 5 · MCQ adaptation

A $\frac{p^6t^{21}}{2000}$

B $\frac{p^6t^{21}}{1000}+1$

C $\frac{p^6t^{21}}{1000}$

D $\frac{p^6t^{21}}{500}$

8.3 MCQ

Simplify. Assume no denominator equals zero.

$$\frac{a^7b^8c^8}{a^5bc^7}$$

Reveal · p. 583 · 6 · MCQ adaptation

A a^2b^7c+1

B a^2b^7c

C $2a^2b^7c$

D $\frac{a^2b^7c}{2}$

8.4 MCQ

Simplify. Assume no denominator equals zero.

$$\left(\frac{3np^3}{7q^2}\right)^2$$

Reveal · p. 583 · 7 · MCQ adaptation

A $\frac{9n^2p^6}{49q^4}$

B $\frac{18n^2p^6}{49q^4}$

C $\frac{9n^2p^6}{98q^4}$

D $\frac{9n^2p^6}{49q^4}+1$

8.6 MCQ

Simplify. Assume no denominator equals zero.

$$\left(\frac{3m^5r^3}{4p^8}\right)^4$$

Reveal · p. 583 · 9 · MCQ adaptation

A $\frac{81m^{20}r^{12}}{512p^{32}}$

B $\frac{81m^{20}r^{12}}{256p^{32}}+1$

C $\frac{81m^{20}r^{12}}{256p^{32}}$

D $\frac{81m^{20}r^{12}}{128p^{32}}$

01 / CHOOSE & CHECK

Q9 — Q10

Q9 Negative and zero exponents
الأسس السالبة والصفرية

Remember For $a \neq 0$, $a^0 = 1$ and $a^{-n} = 1/a^n$. Move only factors with negative exponents.

9.1 MCQ

Simplify using positive exponents. Assume no denominator equals zero and each zero-power base is nonzero.

$$\frac{3m^{-3}t^4p^2}{12t^4}$$

Reveal · p. 591 · 11 · MCQ adaptation

A $-\frac{p^2}{4m^3}$ B $\frac{p^2}{2m^3}$
C $\frac{p^2}{4m^3}$ D 0

9.2 MCQ

Simplify using positive exponents. Assume no denominator equals zero and each zero-power base is nonzero.

$$\left(\frac{3t^6u^2v^5}{9tuv^{21}}\right)^0$$

Reveal · p. 591 · 12 · MCQ adaptation

A 2 B 1
C 0 D -1

9.3 MCQ

Simplify using positive exponents. Assume no denominator equals zero and each zero-power base is nonzero.

$$\frac{x^{-4}y^9}{z^{-2}}$$

Reveal · p. 591 · 13 · MCQ adaptation

A $\frac{y^9z^2}{x^4}$ B 0
C $-\frac{y^9z^2}{x^4}$ D $\frac{2y^9z^2}{x^4}$

9.4 MCQ

Simplify using positive exponents. Assume no denominator equals zero and each zero-power base is nonzero.

$$\left(-\frac{5f^9g^4h^2}{fg^2h^3}\right)^0$$

Reveal · p. 591 · 14 · MCQ adaptation

A 0 B -1
C 2 D 1

Q10 Rational exponents
الأسس النسبية والجذور

Remember a^(1/n) is the principal nth root of a. Evaluate the root before any other power.

10.1 MCQ

Evaluate.

$$\left(\frac{1}{16}\right)^{1/4}$$

Reveal · p. 599 · 9 · MCQ adaptation

A $-\frac{1}{2}$ B $\frac{1}{2}$
C 1 D $\frac{1}{4}$

10.2 MCQ

Evaluate.

$$\sqrt[5]{3125}$$

Reveal · p. 599 · 10 · MCQ adaptation

A 5 B 10
C $\frac{5}{2}$ D -5

10.5 MCQ

Evaluate.

$$\sqrt[6]{4096}$$

Reveal · p. 599 · 13 · MCQ adaptation

A -4 B 4
C 8 D 2

10.6 MCQ

Evaluate.

$$\left(\frac{16}{625}\right)^{1/4}$$

Reveal · p. 599 · 15 · MCQ adaptation

A $\frac{2}{5}$ B $\frac{4}{5}$
C $\frac{1}{5}$ D $-\frac{2}{5}$

01 / CHOOSE & CHECK

Q11

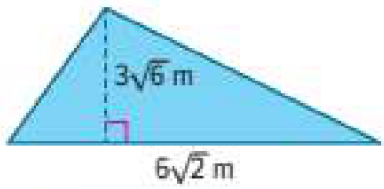
Q11 Products of square roots
ضرب الجذور التربيعية

Remember √a·√b=√(ab) for a,b≥0. Extract perfect-square factors; area uses the full formula.

11.1 MCQ

The base of a triangle is 6√2 m and its height is 3√6 m. What is its area?

Reveal · p. 609 · 67 · MCQ adaptation

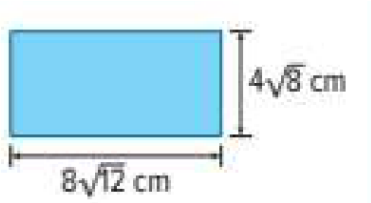


A $18\sqrt{3}$
B $36\sqrt{3}$
C $18\sqrt{2}$
D $9\sqrt{3}$

11.2 MCQ

A rectangle is 8√12 cm long and 4√8 cm wide. What is its area?

Reveal · p. 609 · 68 · MCQ adaptation



A $64\sqrt{6}$
B $32\sqrt{6}$
C $128\sqrt{3}$
D $128\sqrt{6}$

11.3 MCQ

Use √(h·k) to find the geometric mean of 32 and 14 in simplified radical form.

Reveal · p. 609 · 69 · MCQ adaptation

A $16\sqrt{7}$
B $8\sqrt{14}$
C $8\sqrt{7}$
D $4\sqrt{7}$

11.4 MCQ

Fernanda has paint to cover 5 square yards of a cubic prop. Use s=√(A/6). What is the longest side length, to the nearest tenth of a yard?

Reveal · p. 609 · 70 · MCQ adaptation

A $\frac{1}{2}$
B $\frac{9}{10}$
C $\frac{4}{5}$
D $\frac{6}{5}$

01 / CHOOSE & CHECK

Q12

Q12 Quotients of radicals

قسمة الجذور وتبسيطها

Remember √(a/b)=√a/√b for a≥0,b>0. Simplify and rationalise the denominator. For cube roots, extract cubes.

12.1 MCQ

Simplify.

$$\sqrt{\frac{192}{216}}$$

Reveal · p. 607 · 15 · MCQ adaptation

A $\frac{4\sqrt{2}}{3}$
B $\frac{\sqrt{2}}{3}$
C $\frac{2\sqrt{2}}{3}+1$
D $\frac{2\sqrt{2}}{3}$

12.2 MCQ

Simplify.

$$\sqrt{\frac{88}{18}}$$

Reveal · p. 607 · 16 · MCQ adaptation

A $\frac{\sqrt{11}}{3}$
B $1+\frac{2\sqrt{11}}{3}$
C $\frac{2\sqrt{11}}{3}$
D $\frac{4\sqrt{11}}{3}$

12.3 MCQ

Simplify.

$$\sqrt{\frac{84}{121}}$$

Reveal · p. 607 · 17 · MCQ adaptation

A $\frac{2\sqrt{21}}{11}+1$
B $\frac{2\sqrt{21}}{11}$
C $\frac{4\sqrt{21}}{11}$
D $\frac{\sqrt{21}}{11}$

12.4 MCQ

Simplify.

$$\sqrt{\frac{75}{20}}$$

Reveal · p. 607 · 18 · MCQ adaptation

A $\frac{\sqrt{15}}{2}$
B $\sqrt{15}$
C $\frac{\sqrt{15}}{4}$
D $1+\frac{\sqrt{15}}{2}$

01 / CHOOSE & CHECK

Q13 — Q14

Q13 Add and subtract radicals

جمع الجذور وطرحها

Remember Simplify the radicals first; combine coefficients only when their radicands match.

13.1 MCQ

Simplify.

$$7\sqrt{7} - 2\sqrt{7}$$

Reveal · p. 615 · 1 · MCQ adaptation

A $-5\sqrt{7}$ B $1 + 5\sqrt{7}$
C $5\sqrt{7}$ D $10\sqrt{7}$

13.2 MCQ

Simplify.

$$3\sqrt{8} + 7\sqrt{8}$$

Reveal · p. 615 · 2 · MCQ adaptation

A $1 + 20\sqrt{2}$ B $20\sqrt{2}$
C $40\sqrt{2}$ D $-20\sqrt{2}$

13.5 MCQ

Simplify.

$$3\sqrt{5} - 2\sqrt{20}$$

Reveal · p. 615 · 7 · MCQ adaptation

A $\sqrt{5}$ B $1 - \sqrt{5}$
C $-\sqrt{5}$ D $-2\sqrt{5}$

13.6 MCQ

Simplify.

$$3\sqrt{50} - 3\sqrt{32}$$

Reveal · p. 615 · 8 · MCQ adaptation

A $1 + 3\sqrt{2}$ B $3\sqrt{2}$
C $6\sqrt{2}$ D $-3\sqrt{2}$

Q14 Factoring in context

تطبيق التحليل في مسائل تربيعية

Remember Set a physical model equal to the required height/volume, factor, then reject roots outside its domain.

14.1 MCQ

A ballast bag is dropped from 576 ft. Its height is f(t)=−16t²+576. After how many seconds does it hit the ground?

Reveal · p. 722 · 29 · MCQ adaptation

A 36 B 6
C -6 D 18

14.2 MCQ

Factor the volume expression for Catalina's open-topped box.

$$V = 3x^2 - 36x + 108$$

Reveal · p. 722 · 30(a) · MCQ adaptation

A $3(x-6)^2$ B $3(x+6)^2$
C $(3x-6)^2$ D $3(x-12)^2$

14.3 MCQ

Solve by factoring. Check the solutions.

$$x^2 + 9x + 18 = 0$$

Reveal · p. 721 · 15 · MCQ adaptation

A $x = 6, 3$ B $x = -9, -2$
C $x = 0, -9$ D $x = -6, -3$

14.6 MCQ

Solve by factoring. Check the solutions.

$$-x^2 - 12 = -8x$$

Reveal · p. 721 · 18 · MCQ adaptation

A $x = 2, 6$ B $x = -2, -6$
C $x = 3, 4$ D $x = -3, -4$

01 / CHOOSE & CHECK

Q15

Q15 Square root and completing-square methods

الجذر التربيعي وإكمال المربع

Remember Isolate (x−h)²; take both square roots. Complete the square by adding (b/2)².

15.1 MCQ

Which expression does not belong because it is not a perfect square trinomial?

Reveal · p. 730 · 37 · MCQ adaptation

A $n^2 + n + \frac{3}{4}$

B $n^2 - n + \frac{1}{4}$

C $n^2 - \frac{2}{3}n + \frac{1}{9}$

D $n^2 + \frac{1}{3}n + \frac{1}{36}$

15.2 MCQ

Create a quadratic equation for which the only solution is 4. Choose a valid equation.

Reveal · p. 730 · 38 · MCQ adaptation

A $x^2 - 16 = 0$

B $(x + 4)^2 = 0$

C $x^2 - 4 = 0$

D $(x - 4)^2 = 0$

15.3 MCQ

For x²−5x−7=0, which comparison of completing the square, graphing and factoring is correct?

Reveal · p. 730 · 39 · MCQ adaptation

A Graphing always gives exact irrational roots.

B Completing the square gives only one root.

C Completing the square gives exact irrational roots; graphing estimates them; integer factoring does not work.

D Integer factoring gives roots 7 and −1.

15.5 MCQ

Solve by completing the square.

$$x^2 - 2x - 14 = 0$$

Reveal · p. 729 · 11 · MCQ adaptation

A $x = 1 \pm \sqrt{15}$

B $x = -1 \pm \sqrt{15}$

C $x = 1 \pm \sqrt{14}$

D $x = 2 \pm \sqrt{15}$

01 / CHOOSE & CHECK

Q16

Q16 Translations of quadratics
إزاحة الدوال التربيعية

Remember In y=a(x−h)²+k, the vertex is (h,k). x−h shifts right; x+h shifts left.

16.1 MCQ

Describe the translation from the parent function y=x².

$$g(x) = (x-1)^2$$

Reveal · p. 707 · 3 · MCQ adaptation

A Left 1 unit
B Up 1 unit
C Down 1 unit
D Right 1 unit

16.2 MCQ

Describe the translation from the parent function y=x².

$$g(x) = x^2 - 8$$

Reveal · p. 707 · 4 · MCQ adaptation

A Right 8 units
B Left 8 units
C Down 8 units
D Up 8 units

16.5 MCQ

Describe the translation from the parent function y=x².

$$g(x) = (x+5)^2 - 2$$

Reveal · p. 707 · 8 · MCQ adaptation

A Right 5 units and down 2 units
B Left 5 units and up 2 units
C Right 5 units and up 2 units
D Left 5 units and down 2 units

16.6 MCQ

Describe the translation from the parent function y=x².

$$g(x) = -4 + (x-3)^2$$

Reveal · p. 707 · 10 · MCQ adaptation

A Right 3 units and up 4 units
B Left 3 units and up 4 units
C Right 3 units and down 4 units
D Left 3 units and down 4 units

01 / CHOOSE & CHECK

Q17

Q17 Factoring quadratic equations

حل المعادلات التربيعية بالتحليل

Remember Write one side as zero, factor fully, and apply the zero-product property. Check every root.

17.1 MCQ

Solve by factoring. Check the solutions.

$$3b(9b-27)=0$$

Reveal · p. 721 · 5 · MCQ adaptation

A $b=0,-3$
B $b=-3,3$
C $b=0,3$
D $b=3$

17.2 MCQ

Solve by factoring. Check the solutions.

$$(7x+3)(2x-6)=0$$

Reveal · p. 721 · 6 · MCQ adaptation

A $x=-\frac{7}{3},3$
B $x=-\frac{3}{7},3$
C $x=\frac{3}{7},-3$
D $x=-3,3$

17.3 MCQ

Solve by factoring. Check the solutions.

$$b^2=-3b$$

Reveal · p. 721 · 7 · MCQ adaptation

A $b=0,-3$
B $b=-3$
C $b=0,3$
D $b=-3,3$

17.4 MCQ

Solve by factoring. Check the solutions.

$$a^2=4a$$

Reveal · p. 721 · 8 · MCQ adaptation

A $a=4$
B $a=0,-4$
C $a=-4,4$
D $a=0,4$

01 / CHOOSE & CHECK

Q18

Q18 Dilations and reflections of quadratics

تمدد وانعكاس الدوال التربيعية

Remember A negative outside coefficient reflects across the x-axis; |a|>1 stretches vertically. Simplify an inside coefficient before comparing.

18.1 MCQ

Describe the graph relative to the parent y=x².

$$g(x) = -\frac{1}{3}x^2$$

Reveal · p. 707 · 23 · MCQ adaptation

A Reflect across the x-axis; stretch by 4

B Reflect across the x-axis; vertically compress by 1/3

C Reflect across the y-axis; stretch by 4

D Shift down 4 units

18.2 MCQ

Describe the graph relative to the parent y=x².

$$g(x) = \left(-\frac{2}{3}x\right)^2$$

Reveal · p. 707 · 24 · MCQ adaptation

A Vertical compression by 4/9; no reflection

B Reflect across the x-axis; stretch by 9

C Vertical stretch by 3

D Shift left 3 units

18.4 MCQ

Describe the graph relative to the parent y=x².

$$h(x) = -7 - x^2$$

Reveal · p. 708 · 27 · MCQ adaptation

A Stretch by 7, then shift down 1

B Reflect across the x-axis, then shift up 7

C Reflect across the x-axis, then shift down 7

D Reflect across the y-axis, then shift up 7

18.5 MCQ

Describe the graph relative to the parent y=x².

$$g(x) = 2(x-3)^2 + 8$$

Reveal · p. 708 · 28 · MCQ adaptation

A Reflect across the x-axis, then shift up 8

B Stretch vertically by 2, shift right 3 and up 8

C Stretch vertically by 3/2, then shift up 8

D Compress vertically by 2/3, then shift down 8

01 / CHOOSE & CHECK

Q19 — Q20

Q19 Solve quadratics by graphing

حل المعادلات التربيعية بيانيًا

Remember Real solutions are the x-intercepts of the related graph; a tangent gives a repeated root; no intersection means no real solution.

19.1 MCQ

Solve the equation by graphing the related quadratic function.

$$x^2 + 7x + 14 = 0$$

Reveal · p. 713 · 1 · MCQ adaptation

A No real solution
B x = −7, −2
C x = −2, 7
D x = −3.5

19.2 MCQ

Solve the equation by graphing the related quadratic function.

$$x^2 + 2x - 24 = 0$$

Reveal · p. 713 · 2 · MCQ adaptation

A $x = 6, -4$
B $x = -4, 6$
C $x = -12, 2$
D $x = -6, 4$

19.3 MCQ

Solve the equation by graphing the related quadratic function.

$$x^2 + 16x + 64 = 0$$

Reveal · p. 713 · 3 · MCQ adaptation

A $x = -8, 8$
B $x = -16, 4$
C $x = -8$
D $x = 8$

19.6 MCQ

Solve the equation by graphing the related quadratic function.

$$x^2 = 2x - 1$$

Reveal · p. 713 · 6 · MCQ adaptation

A $x = -1$
B $x = -1, 1$
C $x = 2$
D $x = 1$

Q20 Complete the square

إكمال المربع

Remember First make the coefficient of x² equal to 1. Add (b/2)² to form a perfect square.

20.1 MCQ

Find c so that the trinomial is a perfect square.

$$x^2 + 26x + c$$

Reveal · p. 729 · 1 · MCQ adaptation

A 26
B 676
C $\frac{169}{4}$
D 169

20.2 MCQ

Find c so that the trinomial is a perfect square.

$$x^2 - 24x + c$$

Reveal · p. 729 · 2 · MCQ adaptation

A 576
B 36
C 144
D 24

20.3 MCQ

Find c so that the trinomial is a perfect square.

$$x^2 - 19x + c$$

Reveal · p. 729 · 3 · MCQ adaptation

A $\frac{361}{16}$
B $\frac{361}{4}$
C 19
D 361

20.4 MCQ

Find c so that the trinomial is a perfect square.

$$x^2 + 11x + c$$

Reveal · p. 729 · 4 · MCQ adaptation

A $\frac{121}{4}$
B 11
C 121
D $\frac{121}{16}$

02 / WRITE THE STEPS

Q21 · Written practice 1

Q21

Add and subtract polynomials

جمع كثيرات الحدود وطرحها

Remember Distribute a subtraction sign to every term; then collect like terms.

21.1 FRQ · show your working

Find the sum or difference.

$$(3c^3 - c + 11) - (c^2 + 2c + 8)$$

Reveal · p. 636 · 27

(a) Show your simplification.

21.2 FRQ · show your working

Find the sum or difference.

$$(z^2 + z) + (z^2 - 11)$$

Reveal · p. 636 · 28

(a) Show your simplification.

02 / WRITE THE STEPS

Q21 · Written practice 2

Q21

Add and subtract polynomials

جمع كثيرات الحدود وطرحها

Remember Distribute a subtraction sign to every term; then collect like terms.

21.3 FRQ · show your working

Find the sum or difference.

$$(2x - 2y + 1) - (3y + 4x)$$

Reveal · p. 636 · 29

(a) Show your simplification.

21.4 FRQ · show your working

Find the sum or difference.

$$(4a - 5b^2 + 3) + (6 - 2a + 3b^2)$$

Reveal · p. 636 · 30

(a) Show your simplification.

02 / WRITE THE STEPS

Q22 · Written practice 1

Q22

Multiply polynomials

ضرب كثيرات الحدود

Remember Multiply each term in the first polynomial by every term in the second; collect equal powers.

22.1 FRQ · show your working

Find the product.

$$(2y - 11)(y^2 - 3y + 2)$$

Reveal · p. 650 · 18

(a) Show all partial products and combine like terms.

22.2 FRQ · show your working

Find the product.

$$(4a + 7)(9a^2 + 2a - 7)$$

Reveal · p. 650 · 19

(a) Show all partial products and combine like terms.

02 / WRITE THE STEPS

Q22 · Written practice 2

Q22

Multiply polynomials

ضرب كثيرات الحدود

Remember Multiply each term in the first polynomial by every term in the second; collect equal powers.

22.3 FRQ · show your working

Find the product.

$$(m^2 - 5m + 4)(m^2 + 7m - 3)$$

Reveal · p. 650 · 20

(a) Show all partial products and combine like terms.

22.4 FRQ · show your working

Find the product.

$$(x^2 + 5x - 1)(5x^2 - 6x + 1)$$

Reveal · p. 650 · 21

(a) Show all partial products and combine like terms.

02 / WRITE THE STEPS

Q23 · Written practice 1

Q23

Multiplication properties of exponents

خواص ضرب القوى

Remember Add exponents in a product; multiply exponents in a power. Apply the power to the coefficient too.

23.1 FRQ · show your working

A square field has side length $5x^3y^5$. What is its area in terms of x and y?

Reveal · p. 576 · 25

(a) Show the area calculation.

23.2 FRQ · show your working

A cube-shaped shipping box has side length $4m^8n^4$. What is its volume in terms of m and n?



Reveal · p. 576 · 27

(a) Show the volume calculation.

02 / WRITE THE STEPS

Q23 · Written practice 2

Q23

Multiplication properties of exponents

خواص ضرب القوى

Remember Add exponents in a product; multiply exponents in a power. Apply the power to the coefficient too.

23.3 FRQ · show your working

Consider the two expressions.

$$(m^p)(m^q) \quad \text{and} \quad (m^p)^q$$

Reveal · p. 578 · 56

(a) What must be true about p and q for the expressions to be equal? Give an example.

(b) Assuming m, p and q are positive integers, can $(m^p)(m^q) > (m^p)^q$ ever be true? Explain.

23.4 FRQ · show your working

For nonzero real a and b and integers m and t, simplify and explain.

$$\left(-\frac{a^m}{b^t}\right)^{2t}$$

Reveal · p. 578 · 57

(a) Apply the power rules and explain the sign.

02 / WRITE THE STEPS

Q23 · Written practice 3

Q23

Multiplication properties of exponents

خواص ضرب القوى

Remember Add exponents in a product; multiply exponents in a power. Apply the power to the coefficient too.

23.5 FRQ · show your working

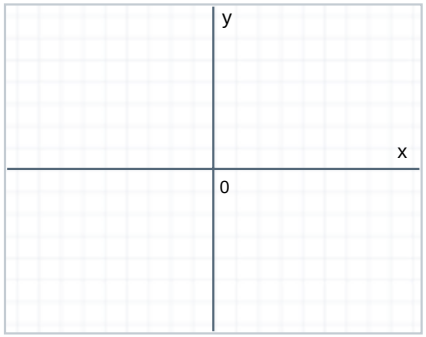
Consider y=x, y=x² and y=x³.

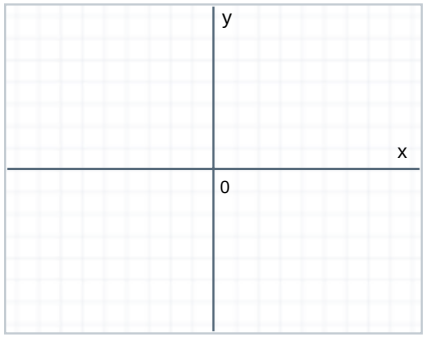
Reveal · p. 578 · 58

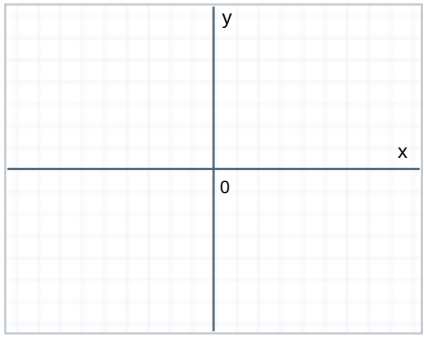
Equation	Related expression	Power of x	Linear / nonlinear
$y = x$			
$y = x^2$			
$y = x^3$			

(a) For each equation, write the related expression and record the power of x.

(b) Graph each equation using a graphing calculator.







(c) Classify each graph as linear or nonlinear.

(d) Explain how to determine whether an equation, or its related expression, is linear or nonlinear without graphing.

02 / WRITE THE STEPS

Q24 · Written practice 1

Q24

Quadratic extrema and graph features

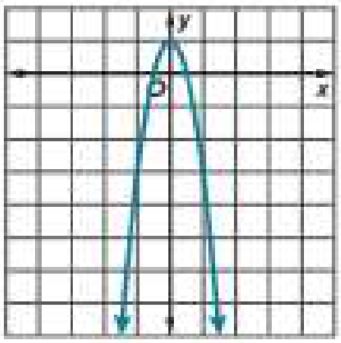
القيم العظمى والصغرى وخصائص المنحنى

Remember The vertex (h,k) gives the minimum k when the graph opens up and maximum k when it opens down; the axis is x=h.

24.1 FRQ · show your working

Identify the axis of symmetry, vertex and y-intercept of the graph. Then describe the end behavior.

Reveal · p. 695 · 1



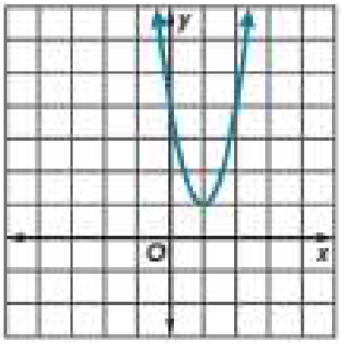
(a) Axis of symmetry and vertex.

(b) y-intercept and end behavior.

24.2 FRQ · show your working

Identify the axis of symmetry, vertex and y-intercept of the graph. Then describe the end behavior.

Reveal · p. 695 · 2



(a) Axis of symmetry and vertex.

(b) y-intercept and end behavior.

02 / WRITE THE STEPS

Q24 · Written practice 2

Q24

Quadratic extrema and graph features

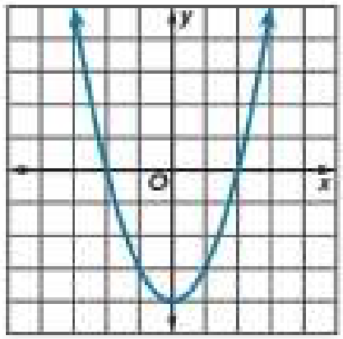
القيم العظمى والصغرى وخصائص المنحنى

Remember The vertex (h,k) gives the minimum k when the graph opens up and maximum k when it opens down; the axis is x=h.

24.3 FRQ · show your working

Identify the axis of symmetry, vertex and y-intercept of the graph. Then describe the end behavior.

Reveal · p. 695 · 3



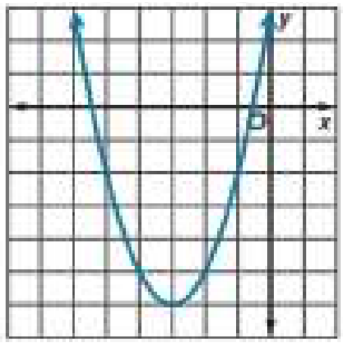
(a) Axis of symmetry and vertex.

(b) y-intercept and end behavior.

24.4 FRQ · show your working

Identify the axis of symmetry, vertex and y-intercept of the graph. Then describe the end behavior.

Reveal · p. 695 · 4



(a) Axis of symmetry and vertex.

(b) y-intercept and end behavior.

02 / WRITE THE STEPS

Q25 · Written practice 1

Q25

Write a quadratic from its graph

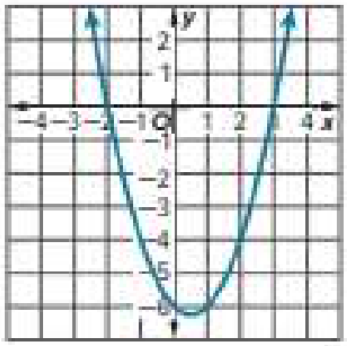
كتابة الدالة التربيعية من الرسم

Remember Read the zeros r,s, write $y=a(x-r)(x-s)$, then use another point to find a. Expand to $y=ax^2+bx+c$.

25.1 FRQ · show your working

Write a quadratic function for the given graph. Give the equation in standard form.

Reveal · p. 722 · 31

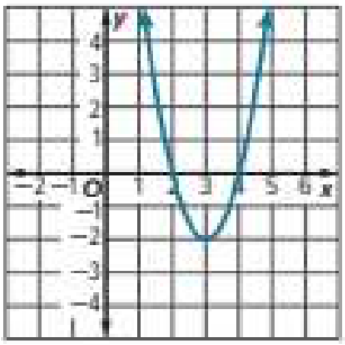


(a) Show how you find the equation.

25.2 FRQ · show your working

Write a quadratic function for the given graph. Give the equation in standard form.

Reveal · p. 722 · 32



(a) Show how you find the equation.

02 / WRITE THE STEPS

Q25 · Written practice 2

Q25 **Write a quadratic from its graph**
كتابة الدالة التربيعية من الرسم

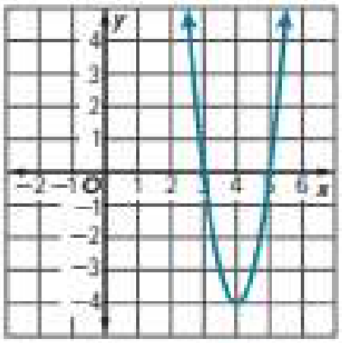
> **Remember** Read the zeros r,s, write y=a(x−r)(x−s), then use another point to find a. Expand to y=ax²+bx+c.

25.3 FRQ · show your working

Write a quadratic function for the given graph. Give the equation in standard form.

Reveal · p. 722 · 33

(a) Show how you find the equation.

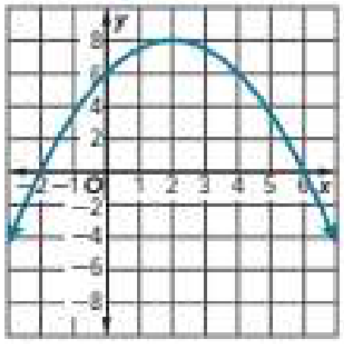


25.4 FRQ · show your working

Write a quadratic function for the given graph. Give the equation in standard form.

Reveal · p. 722 · 34

(a) Show how you find the equation.



03 / CHECK & LEARN

MCQ answer key

Check after solving. Match the printed question number to its correct option. Full explanations are available in the online LO bank.

Q1 · FOIL binomial products
1.1: C 1.2: B 1.3: A 1.4: D

Q2 · Squares of sums and differences
2.1: B 2.2: A 2.3: D 2.4: C

Q3 · Sum and difference products
3.1: A 3.2: D 3.3: C 3.4: B

Q4 · Factor using distribution
4.1: D 4.2: C 4.3: B 4.4: A

Q5 · Analyse monic trinomial factors
5.2: B 5.3: A 5.5: C 5.6: B

Q6 · Factor non-monic trinomials
6.1: B 6.2: A 6.3: D 6.4: C

Q7 · Difference of squares
7.1: A 7.2: D 7.3: C 7.4: B

Q8 · Divide monomials
8.2: C 8.3: B 8.4: A 8.6: C

Q9 · Negative and zero exponents
9.1: C 9.2: B 9.3: A 9.4: D

Q10 · Rational exponents
10.1: B 10.2: A 10.5: B 10.6: A

Q11 · Products of square roots
11.1: A 11.2: D 11.3: C 11.4: B

Q12 · Quotients of radicals
12.1: D 12.2: C 12.3: B 12.4: A

Q13 · Add and subtract radicals
13.1: C 13.2: B 13.5: C 13.6: B

Q14 · Factoring in context
14.1: B 14.2: A 14.3: D 14.6: A

Q15 · Square root and completing-square methods
15.1: A 15.2: D 15.3: C 15.5: A

Q16 · Translations of quadratics
16.1: D 16.2: C 16.5: D 16.6: C

Q17 · Factoring quadratic equations
17.1: C 17.2: B 17.3: A 17.4: D

Q18 · Dilations and reflections of quadratics
18.1: B 18.2: A 18.4: C 18.5: B

Q19 · Solve quadratics by graphing
19.1: A 19.2: D 19.3: C 19.6: D

Q20 · Complete the square
20.1: D 20.2: C 20.3: B 20.4: A

03 / CHECK & LEARN

Worked solutions 1

21.1 p. 636 · 27

(a) Correct sign distribution and like-term collection (6 marks); simplified polynomial 3·c³ - c² - 3·c + 3 (4 marks).

$$3c^3 - c^2 - 3c + 3$$

21.2 p. 636 · 28

(a) Correct sign distribution and like-term collection (6 marks); simplified polynomial 2·z² + z - 11 (4 marks).

$$2z^2 + z - 11$$

21.3 p. 636 · 29

(a) Correct sign distribution and like-term collection (6 marks); simplified polynomial -2·x - 5·y + 1 (4 marks).

$$-2x - 5y + 1$$

21.4 p. 636 · 30

(a) Correct sign distribution and like-term collection (6 marks); simplified polynomial 2·a - 2·b² + 9 (4 marks).

$$2a - 2b^2 + 9$$

22.1 p. 650 · 18

(a) All partial products (6 marks); correct collection of like terms (4 marks): 2·y³ - 17·y² + 37·y - 22.

$$2y^3 - 17y^2 + 37y - 22$$

22.2 p. 650 · 19

(a) All partial products (6 marks); correct collection of like terms (4 marks): 36·a³ + 71·a² - 14·a - 49.

$$36a^3 + 71a^2 - 14a - 49$$

22.3 p. 650 · 20

(a) All partial products (6 marks); correct collection of like terms (4 marks): m⁴ + 2·m³ - 34·m² + 43·m - 12.

$$m^4 + 2m^3 - 34m^2 + 43m - 12$$

03 / CHECK & LEARN

Worked solutions 2

22.4 p. 650 · 21

(a) All partial products (6 marks); correct collection of like terms (4 marks): 5·x⁴ + 19·x³ - 34·x² + 11·x - 1.

$$5x^4 + 19x^3 - 34x^2 + 11x - 1$$

23.1 p. 576 · 25

(a) A=s²=(5x³y⁵)²=25x⁶y¹⁰. Square the coefficient (3), multiply each exponent by 2 (5), final expression (2).

$$A = 25x^6y^{10}$$

23.2 p. 576 · 27

(a) V=s³=(4m⁸n⁴)³=64m²⁴n¹². Cube the coefficient (3), multiply each exponent by 3 (5), final expression (2).

$$V = 64m^{24}n^{12}$$

23.3 p. 578 · 56

(a) For positive m≠1, require p+q=pq, or (p−1)(q−1)=1 (3). Example p=q=2 (2). Without that assumption, m=1 makes both expressions equal for all defined p,q; other bases require domain care.

$$p + q = pq \quad (m > 0,\ m \neq 1)$$

(b) Yes. Take m=2 and p=q=1: 2·2=4>2¹=2 (3). For m>1 compare exponents p+q and pq (2).

$$2^1 \cdot 2^1 = 4 > 2^{1\cdot 1} = 2$$

23.4 p. 578 · 57

(a) Since 2t is even, (−1)²ᵗ=1 (3). Multiply the numerator exponents to get 2mt (3), and denominator exponents to get 2t² (3); final quotient (1). Nonzero a,b make all integer powers defined.

$$\frac{a^{2mt}}{b^{2t^2}}$$

23.5 p. 578 · 58

(a) x=x¹ (power 1), x·x=x² (power 2), x·x·x=x³ (power 3). One mark per row.

(b) Correct line, upward parabola and increasing cubic through the origin. One mark per graph.

(c) y=x is linear; y=x² and y=x³ are nonlinear. One mark for all three correct.

(d) After simplification, y=ax+b is linear (2); higher powers of x such as x² and x³ are nonlinear (1).

24.1 p. 695 · 1

(a) Axis x=0 (2); vertex (0,1) (3).

(b) y-intercept (0,1) (2); y→−∞ as x→±∞; maximum 1 (3). The extremum is the vertex y-coordinate.

03 / CHECK & LEARN

Worked solutions 3

24.2 p. 695 · 2

(a) Axis x=1 (2); vertex (1,1) (3).

(b) y-intercept (0,4) (2); y → +∞ as x → ±∞; minimum 1 (3). The extremum is the vertex y-coordinate.

24.3 p. 695 · 3

(a) Axis x=0 (2); vertex (0,−4) (3).

(b) y-intercept (0,−4) (2); y → +∞ as x → ±∞; minimum −4 (3). The extremum is the vertex y-coordinate.

24.4 p. 695 · 4

(a) Axis x=−3 (2); vertex (−3,−6) (3).

(b) y-intercept (0,3) (2); y → +∞ as x → ±∞; minimum −6 (3). The extremum is the vertex y-coordinate.

25.1 p. 722 · 31

(a) Zeros −2 and 3 give y=a(x+2)(x−3). The y-intercept −6 gives a=1. Correct reading of two graph features (4), coefficient a (3), expanded standard form (3).

$$y = x^2 - x - 6$$

25.2 p. 722 · 32

(a) Zeros 2 and 4 give y=a(x−2)(x−4). The vertex (3,−2) gives a=2. Correct reading of two graph features (4), coefficient a (3), expanded standard form (3).

$$y = 2x^2 - 12x + 16$$

25.3 p. 722 · 33

(a) Zeros 3 and 5 give y=a(x−3)(x−5). The vertex (4,−4) gives a=4. Correct reading of two graph features (4), coefficient a (3), expanded standard form (3).

$$y = 4x^2 - 32x + 60$$

25.4 p. 722 · 34

(a) Zeros −2 and 6 give y=a(x+2)(x−6). The vertex (2,8) gives a=−½. Correct reading of two graph features (4), coefficient a (3), expanded standard form (3).

$$y = -\tfrac{1}{2}x^2 + 2x + 6$$